

Appendix

PROOF. Let $\mathcal{M} = (S, P)$ be the (unknown) DTMC governing the agent's behavior, and let $\hat{\mathcal{M}} = (S, \hat{P}_\Pi^\alpha)$ be the DTMC returned by Algorithm 1. Fix a CTL property ψ , accuracy ε , and confidence δ .

Step 1: Transition-probability concentration. For each state $s \in S$, the algorithm maintains the weighted visitation count n_s^Π . Define the state-wise threshold

$$H^*(n_s^\Pi, \varepsilon', \delta') = \max_{t \in S} H(n_s^\Pi, n_{st}^\Pi, \varepsilon', \delta'), \quad \delta' = \frac{\delta}{|S|},$$

where $H(\cdot)$ is the transition-wise concentration bound of Theorem 6. By the stopping rule of Algorithm 1, when the algorithm terminates,

$$n_s^\Pi \geq \left(\frac{11}{10} B(\hat{P}_\Pi^\alpha) \right)^2 H^*(n_s^\Pi, \varepsilon, \frac{\delta}{|S|}), \quad \forall s \in S.$$

Thus, by Theorem 6, with probability at least $1 - \delta'$,

$$\max_{t \in S} |P(s, t) - \hat{P}_\Pi^\alpha(s, t)| \leq \varepsilon', \quad \varepsilon' = \frac{\varepsilon}{B(\hat{P}_\Pi^\alpha)}.$$

Applying a union bound over all $s \in S$ gives

$$\Pr \left(\max_{s, t \in S} |P(s, t) - \hat{P}_\Pi^\alpha(s, t)| \leq \varepsilon' \right) \geq 1 - \delta. \quad (4)$$

Step 2: Propagating local error to CTL satisfaction probabilities. By Theorem 5 (reachability sensitivity), for any CTL property ψ ,

$$|\mathbb{P}_\mathcal{M}(\psi) - \mathbb{P}_{\hat{\mathcal{M}}}(\psi)| \leq B(\hat{P}_\Pi^\alpha) \cdot \max_{s, t \in S} |P(s, t) - \hat{P}_\Pi^\alpha(s, t)|.$$

Under the good event of Eq. (4),

$$\max_{s, t} |P(s, t) - \hat{P}_\Pi^\alpha(s, t)| \leq \varepsilon' = \frac{\varepsilon}{B(\hat{P}_\Pi^\alpha)},$$

implying

$$|\mathbb{P}_\mathcal{M}(\psi) - \mathbb{P}_{\hat{\mathcal{M}}}(\psi)| \leq B(\hat{P}_\Pi^\alpha) \varepsilon' = \varepsilon.$$

Step 3: PAC guarantee. Combining the above yields

$$\Pr \left(|\mathbb{P}_{\hat{\mathcal{M}}}(\psi) - \mathbb{P}_\mathcal{M}(\psi)| \leq \varepsilon \right) \geq 1 - \delta,$$

which is precisely (ε, δ) -PAC correctness (Definition 3.3). Thus Algorithm 1 is PAC-correct. $\square$